

Assignment 6 Solutions

Modern Cryptography (2)

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Problem 1

1. The syndrome of a single-bit error vector e_i is exactly the i -th column of H . Hence

$$s(e_1) = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad s(e_2) = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad s(e_3) = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix},$$

$$s(e_4) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad s(e_5) = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

2. We compute

$$Hr^T = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

This is the syndrome of e_1 , so the error is in the first position. Thus

$$e = e_1 = (1, 0, 0, 0, 0),$$

and the decoded codeword is

$$c = r - e = (1, 0, 1, 1, 1) + (1, 0, 0, 0, 0) = (0, 0, 1, 1, 1),$$

where the addition is over $\mathbb{F}_2$.

Problem 2

1. Since $u = (u_1, u_2) \in \mathbb{F}_2^2$, the codewords are

$$00 \cdot G = (0, 0, 0, 0, 0),$$

$$10 \cdot G = (1, 0, 1, 1, 0),$$

$$01 \cdot G = (0, 1, 1, 0, 1),$$

$$11 \cdot G = (1, 1, 0, 1, 1).$$

Hence

$$C = \{00000, 10110, 01101, 11011\}.$$

2. The code length is $n = 5$, the dimension is $k = 2$, and the nonzero codewords have weights

$$\text{wt}(10110) = 3, \quad \text{wt}(01101) = 3, \quad \text{wt}(11011) = 4.$$

Hence

$$d = 3,$$

so the code has parameters

$$[5, 2, 3].$$

3. Since

$$G = (I_2 \mid P), \quad P = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix},$$

one parity-check matrix is

$$H = (P^T \mid I_3) = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix}.$$

4. For

$$r = (1, 1, 0, 0, 1),$$

we compute

$$Hr^T = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

This is the fourth column of H , so the smallest-weight coset leader is

$$e = (0, 0, 0, 1, 0).$$

Hence the decoded codeword is

$$c = r - e = (1, 1, 0, 1, 1).$$

Problem 3

1. Let

$$f(x) = ax^2 + bx + c.$$

Using the first three coordinates,

$$f(0) = 2, \quad f(1) = 6, \quad f(2) = 5.$$

From $f(0) = 2$, we get

$$c = 2.$$

Then

$$a + b + 2 = 6 \quad \Rightarrow \quad a + b = 4,$$

and

$$4a + 2b + 2 = 5 \quad \Rightarrow \quad 4a + 2b = 3.$$

Dividing the second equation by 2 in $\mathbb{F}_7$ gives

$$2a + b = 5.$$

Subtracting $a + b = 4$ from this, we obtain

$$a = 1,$$

and hence

$$b = 3.$$

Therefore

$$f(x) = x^2 + 3x + 2.$$

2. The corresponding codeword is

$$(f(0), f(1), f(2), f(3), f(4), f(5)) = (2, 6, 5, 6, 2, 0).$$

Comparing with

$$r = (2, 6, 5, 6, 2, 6),$$

we see that the only error is at the sixth coordinate, corresponding to the evaluation point $x = 5$. The error value is

$$6 - 0 = 6 \in \mathbb{F}_7.$$

3. This is an $[6, 3, 4]$ Reed–Solomon code, so its error-correcting capability is

$$t = \left\lfloor \frac{4 - 1}{2} \right\rfloor = 1.$$

The Hamming distance between the received word and the decoded codeword is exactly

$$1,$$

which is within the decoding radius.

Problem 4

1. The Berlekamp–Welch equations are

$$E(\alpha_i)y_i = N(\alpha_i), \quad i = 1, \dots, 5.$$

Since

$$E(x) = e_1x + e_0, \quad N(x) = a_0 + a_1x + a_2x^2,$$

we obtain:

for $\alpha_1 = 0$,

$$e_0 \cdot 1 = a_0;$$

for $\alpha_2 = 1$,

$$(e_1 + e_0) \cdot 3 = a_0 + a_1 + a_2;$$

for $\alpha_3 = 2$,

$$(2e_1 + e_0) \cdot 0 = a_0 + 2a_1 + 4a_2;$$

for $\alpha_4 = 3$,

$$(3e_1 + e_0) \cdot 0 = a_0 + 3a_1 + 2a_2;$$

for $\alpha_5 = 4$,

$$(4e_1 + e_0) \cdot 2 = a_0 + 4a_1 + 2a_2.$$

Equivalently,

$$\begin{aligned}a_0 &= e_0, \\a_0 + a_1 + a_2 &= 3e_1 + 3e_0, \\a_0 + 2a_1 + 4a_2 &= 0, \\a_0 + 3a_1 + 2a_2 &= 0, \\a_0 + 4a_1 + 2a_2 &= e_1 + 2e_0.\end{aligned}$$

2. Solving over $\mathbb{F}_7$, we obtain

$$(e_1, e_0, a_0, a_1, a_2) = \lambda(1, 5, 5, 4, 2)$$

for some $\lambda \in \mathbb{F}_7^\times$.

Hence all nonzero solutions are of the form

$$E(x) = \lambda(x + 5), \quad N(x) = \lambda(2x^2 + 4x + 5), \quad \lambda \in \mathbb{F}_7^\times.$$

3. Therefore

$$f(x) = \frac{N(x)}{E(x)} = \frac{\lambda(2x^2 + 4x + 5)}{\lambda(x + 5)} = \frac{2x^2 + 4x + 5}{x + 5} = 2x + 1.$$

So the transmitted polynomial is

$$f(x) = 2x + 1.$$

4. Since one may choose

$$E(x) = x + 5 = x - 2,$$

the error occurred at $x = 2$, i.e. at the third coordinate.

The transmitted codeword is

$$(f(0), f(1), f(2), f(3), f(4)) = (1, 3, 5, 0, 2),$$

while the received word is

$$(1, 3, 0, 0, 2).$$

Hence the error value is

$$0 - 5 = -5 \equiv 2 \pmod{7}.$$